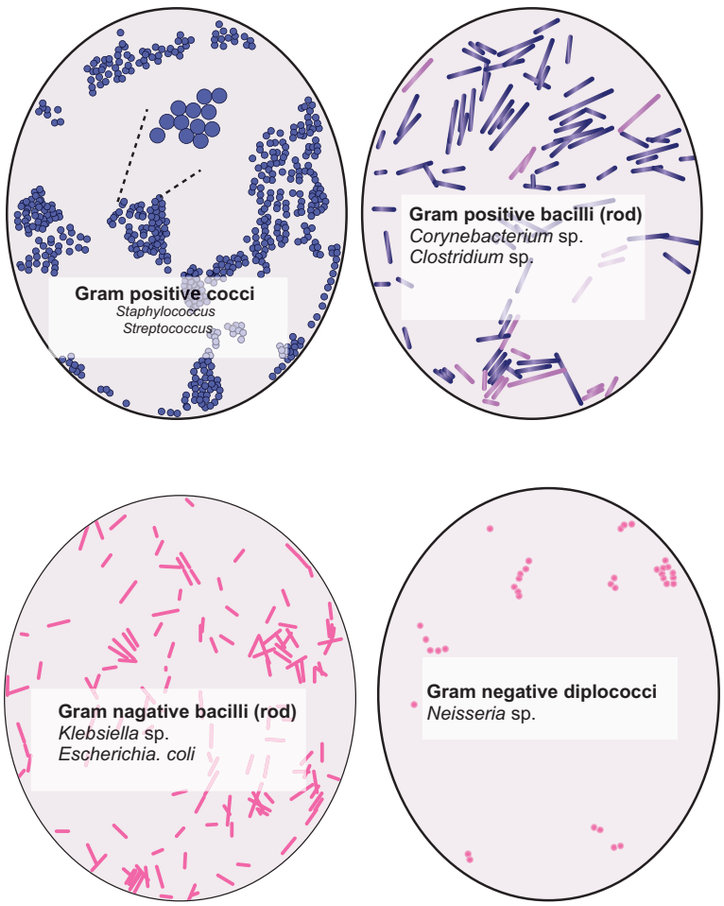
Common Bacteria with their Gram reactions



3.9 Commonly used Stains and its Applications

Lactophenol cotton blue stain is the most widely used for staining and observing fungi. Giemsa stain is a Romanowsky stain, widely used in microbiology laboratory for staining of blood and blood parasites like malarial protozoans. Calcofluor white stain is commonly used stain to directly detect the fungal elements in tissues and in culture.

Acridine orange stain is used to confirm the presence of bacteria in blood cultures when Gram stain results are difficult to interpret using light microscopy. The stain binds to nucleic acid and stains them. It is also used for the detection of cell wall deficient bacteria example Mycoplasma. Fluorochrome stains such as auramine-rhodamine stains are readily available to detect the bacteria in the specimens through Fluorescent microscopy.

Summary

Staining makes microscopic semi transparent bacterial cell visible. It is a substance that adheres to a cell and impart colour. On the basis of the chemical composition, stains or dyes are classified as acidic, basic and neutral. Staining techniques are classified as simple, differential and special. Simple staining uses a single dye and can help to identify the shape and size of an organism. Differential staining use more than one dye to distinguish between structures in a cell or different types of cells. The Gram stain procedure divides bacteria into Gram positive and Gram negative bacteria. Specialized staining such as endospore staining is used to detect the presence of endospores in bacteria.

TN_GOVT_XI_Micro_Biology_CH03.indd 37 16-02-2019 9.55.52 AM